AI 时代商业工程 · 课题研究系列
BUSINESS ENGINEERING IN THE AI ERA · RESEARCH STUDY SERIES

No. 03 / 12

课题研究报告 · RESEARCH STUDY

代理式组织：从个体生产率到大战略

The Agentic Organization: From Individual Productivity to Grand Strategy

关于中性收益、治理中层与十年瞄准的课题研究

A research study on neutral gains, the governed middle and the decade aim

研究底本 SOURCE
Cuofano, G. The Agentic Organization (2026)

版本 EDITION
2026 年 10 月 · October 2026

语言 LANGUAGE
中英双语 · Bilingual Chinese–English

摘要 *Abstract*

AI 工具显著提升了个体生产率，但许多企业发现，仪表盘上的收益并未转化为利润率、赢单率或定价能力。本研究以 Cuofano《代理式组织》（2026）为底本，分析这一“中性收益”现象，构建由个体、群体、组织与战略组成的**四尺度框架**，并把底本自身的证伪条件改写为五项假设。研究发现：其一，个体增益偏重新手、边界参差且完全对称，因而被竞争传导为市场基准线；其二，群体生产率是被设计的而非相加的，混合团队通过按回路切分工作、保护裁决者席位并经验性地设定比例打破中性，同时产出可复利的判断记录；其三，**治理中层**是把边缘战术转化为标准与方向的传动装置，而在战术充裕的时代，护城河由四块组织性基石构成。对 Cursor、Shopify、JPMorgan 等公开案例及一个复合案例的比较表明，领先者并非买了不同的工具，而是率先完成了重组。由于证据以任务和季度为尺度，战略层结论仍属外推；文末提出研究议程。

AI tools have lifted individual productivity markedly, yet many firms find that dashboard gains do not become better margin, win rate or pricing power. Taking Cuofano's *The Agentic Organization* (2026) as its source, this study analyses this neutral-gain phenomenon, builds a **four-scale framework** of individual, group, organization and strategy, and restates the source's own falsifiers as five hypotheses. Three findings emerge. First, the individual gain is bottom-heavy, jagged and fully symmetric, so competition passes it into the market baseline. Second, group productivity is designed, not added: mixed teams break neutrality by re-dividing work into loops, protecting the adjudicator's seat and setting ratios empirically, while producing a compounding record of judgment. Third, the **governed middle** is the transmission that converts edge tactics into standards and direction, and in an era of abundant tactics the moat consists of four organizational stones. A comparison of public cases such as Cursor, Shopify and JPMorgan, plus a composite case, indicates that leaders did not buy different tools but reorganized first. As the evidence spans tasks and quarters, strategy-level conclusions remain extrapolations; a research agenda follows.

关键词 代理式组织；中性收益；人机混合团队；治理中层；残留记录；大战略

KEYWORDS agentic organization; neutral gain; mixed human-agent teams; governed middle; decision residue; grand strategy

研究问题 *Research Questions*

RQ1 为什么对称可得的 AI 工具带来的个体生产率提升，在企业层面趋于中性？

Why are individual gains from symmetrically available AI tools neutral at firm level?

RQ2 群体层面需要怎样的设计，才能把人与代理组合为超越个体加总的系统？

What group design makes humans and agents exceed the sum of individuals?

RQ3 组织如何协调自下而上的发现与自上而下的方向，把分散战术转化为企业资产？

How can bottom-up discovery and top-down direction turn scattered tactics into firm assets?

RQ4 在执行变得廉价的时代，什么构成可防御的竞争位置？

When execution is cheap, what makes a position defensible?

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第 1 章 · CHAPTER 1

01 研究背景与问题提出

Background and Problem Statement

当每个竞争者都能以相同价格买到相同的智能，个体生产率的提升为何没有转化为企业优势？这是本研究的起点。

When every competitor can buy the same intelligence at the same price, why do individual productivity gains fail to become firm advantage? That puzzle is where this study begins.

中文

ENGLISH

1.1 “绿色仪表盘”悖论

The Green-Dashboard Paradox

底本以一个刻意构造的场景开篇：一家公司全面采用生成式工具后，工程交付量提升约四成，市场内容产出增至五倍，分析周期由一周缩短到数小时；每个团队都能证明自己的收益。然而决定竞争结果的指标——利润率、赢单率、实现价格——要么纹丝不动，要么悄然下滑。主要对手在同一季度以同样的新速度推出了同样的功能。

The source volume opens with a deliberately constructed scene. After a company adopts generative tools across the board, engineering ships roughly forty percent more, marketing produces five times the content, and analysis that took a week now takes hours; every team can prove its gains. Yet the numbers that decide competition, margin, win rate and realised price, either stay flat or drift down. The main rivals shipped the same features in the same quarter at the same new speed.

+40%
工程交付量
engineering output

5×
内容产出
content output

±0
利润率与赢单率
margin and win rate

示意 Illustrative: 底本开篇情景中的数字，非实测 figures from the source's opening vignette, not measurements. Cuofano (2026).

这一悖论并不意味着收益是虚假的。每一项个体测量都是诚实的，收益也没有消失，而是流向了一个确定的去处：市场的新基准线。客户把更快的交付和更好的产出当作新常态，企业则以更快的速度维持原有的相对位置。底本将此概括为一个核心命题：**任何竞争者都能购买的工具所带来的生产率增益，在个人层面是真实的，在企业层面是中性的。**

The paradox does not mean the gains were fictitious. Every individual measurement was honest, and the gains did not vanish; they went somewhere specific, into the market's new baseline. Customers absorb faster delivery and better output as the new normal, and firms keep running, faster, in the same relative positions. The source condenses this into its founding proposition: **a productivity gain from a tool every competitor can buy is real at the level of the person and neutral at the level of the firm.**

> **运营效率不等于战略。**
>
> *Operational effectiveness is not strategy.*
>
> — Michael E. Porter, “What Is Strategy?”, HBR, 1996

1.2 地基的两次位移
Two Shifts in the Ground

问题之所以在此时变得尖锐，是因为企业所处的地基同时在两个方向上移动。第一是**市场面**：企业面对的网络正在获得第二类访客。研究品类、比较报价、核查集成乃至完成购买的，越来越多是代表人类行事的软件代理。为说服人眼而建的门面，对一个读取结构、核验声明、不会被打动的机器买家而言几乎不可读。

The problem has sharpened now because the ground beneath the firm moved in two directions at once. The first is the **market face**: the web the firm faces is acquiring a second population. Agents acting for people increasingly research the category, compare offers, check integrations and, where the rails exist, complete the purchase. A storefront built to persuade the human eye is largely illegible to a machine buyer that reads structure, verifies claims and cannot be charmed.

第二是**工具面**：软件正从员工操作的工具变为直接交付工作的“对手方”，底本称之为代理式工作即服务（AGaaS）。工具放大一个人，组织图不变；对手方替代一项任务，组织图就在某处失效了。供应商随之从按席位计价转向按请求、按解决、按结果计价，而企业内部的核算仍以“人”为单位。

The second is the **tooling face**: software is turning from an instrument an employee operates into a counterparty that delivers work, which the source calls agentic work as a service (AGaaS). A tool amplifies a person and the org chart stays put; a counterparty replaces a task and the org chart is now wrong somewhere. Vendors accordingly reprice from seats toward requests, resolutions and outcomes, while the firm's internal accounting still runs in units of people.

1.3 问题提出
Problem Statement

由此，本研究的核心问题不是“工具是否有效”，而是“收益在被个人生产出来之后去了哪里”。底本给出的答案是三个动词：在群体层面被**组织**，在组织层面被**治理**，在战略层面被**瞄准**。大多数转型停留在第一层就开始庆祝；本研究追问的是向上攀升的机制、条件与可检验性。

The core question of this study is therefore not whether the tools work, but where the gain goes after the individual produces it. The source answers with three verbs: the gain must be **organized** at the group level, **governed** at the organization level and **aimed** at the level of strategy. Most transformations stop at the first altitude and celebrate; this study examines the mechanisms, conditions and testability of the climb.

图 1-1 从中性收益到可防御位置的攀升路径
Figure 1-1 *The climb from neutral gain to defended position*



第 2 章 · CHAPTER 2

02 核心概念与理论框架

Core Concepts and Theoretical Framework

本章把底本的论证整理为一个“四尺度”分析框架，并界定贯穿全文的七个分析工具。

This chapter organizes the source's argument into a four-scale analytical framework and defines the seven instruments used throughout.

中文　　ENGLISH

2.1 四尺度框架

The Four-Scale Framework

底本把企业的智能化转型拆分为四个尺度：个体、群体、组织与战略。其关键主张是，价值并不在单一尺度内生成，而是在尺度之间的“转换”中形成或流失。个体尺度生产原材料；群体尺度通过设计把原材料组合为系统；组织尺度把分散的战术转化为标准与方向；战略尺度则把这一切指向一个对手在一个季度内无法复制的位置。

The source decomposes the firm's transformation into four scales: the individual, the group, the organization and the strategy. Its key claim is that value is not created within any single scale but is formed or lost in the conversions between them. The individual scale produces raw material; the group scale composes it by design into systems; the organization scale converts scattered tactics into standards and direction; and the strategy scale aims all of it at a position a rival cannot copy in a quarter.

图 2-1 四尺度分析框架
Figure 2-1 *The four-scale analytical framework*

战略 Strategy	在战术充裕时进行选择；以十年为期界定位置并重建护城河	Choose amid abundant tactics; define a decade position and rebuild the moat
组织 Organization	治理中层：把边缘发现转化为标准，再转化为方向	The governed middle: discoveries into standards, standards into direction
群体 Group	人机混合团队与代理群体；生产率是被设计的而非相加的	Mixed teams and swarms; productivity is designed, not added
个体 Individual	真实但参差、对称的增益；是入场费而非奖品	Real, jagged, symmetric lift; the entry fee, not the prize

来源 Source: 根据 Cuofano (2026) 的章节结构整理 organized from the source's chapter structure.

2.2 历史参照：通用技术的组织滞后

Historical Anchor: the Organizational Lag of General-Purpose Technologies

框架的历史依据来自经济史。十九世纪末工厂电气化时，工程师只是用发电机替换了蒸汽机，厂房仍围绕主传动轴而非工作流排布。按 Paul David 的研究，测得的生产率在约四十年间几乎没有变化；直到小型电机让主轴得以拆除、厂房按物料流重新布局，收益才到来。

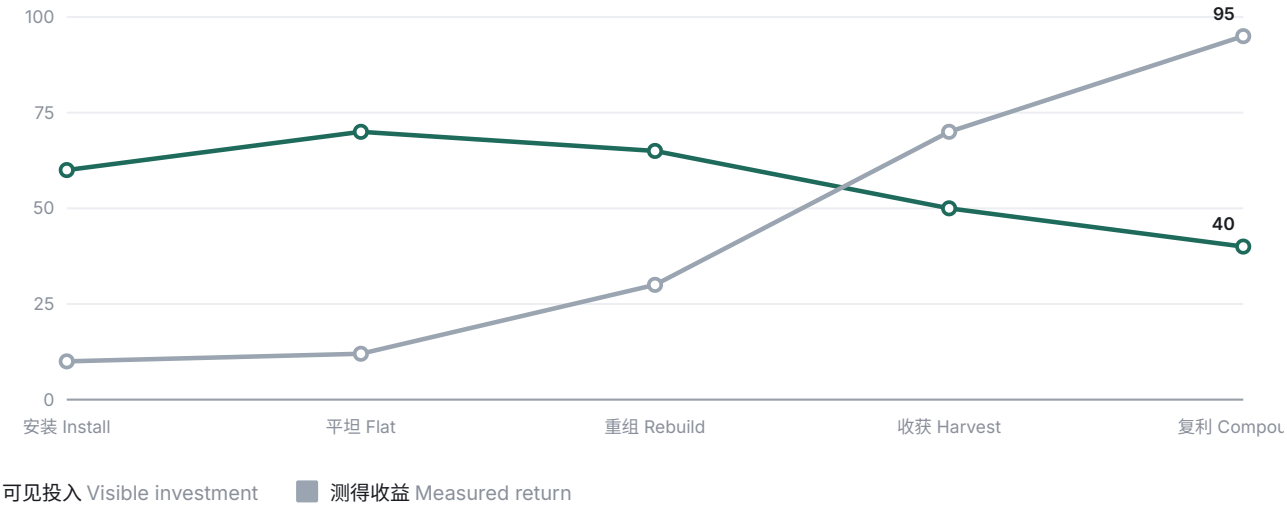
The framework's historical grounding comes from economic history. When factories electrified in the late nineteenth century, engineers simply swapped the steam engine for a dynamo, leaving the building arranged around the line shaft rather than the flow of work. As Paul David documented, measured productivity barely moved for about forty years; the

gain arrived only when small motors let the shaft come out and factories were laid out along materials flow.

计算机在一个世纪后重演了同样的“J 曲线”（Brynjolfsson、Rock 与 Syverson）。底本据此把今天的“主传动轴”命名为：**围绕稀缺人类注意力构建的工作流架构**——按一个人的一天切分岗位，按一个经理的管理幅度设计审批链，按一个人的日历安排会议。

Computing repeated the curve a century later, formalized as the productivity J-curve by Brynjolfsson, Rock and Syverson. On that basis the source names today's line shaft: **the workflow architecture built around scarce human attention**, with jobs sized to one person's day, approval chains sized to one manager's span and meetings sized to one calendar.

图 2-2 通用技术的 J 曲线：投入与测得收益（示意）
Figure 2-2 *The general-purpose-technology J-curve: investment and measured returns (illustrative)*



示意 Illustrative: 曲线形状依据底本对 David (1990) 与 Brynjolfsson 等 (2021) 的转述，数值无实测含义 shape follows the source's account; values carry no measured meaning.

> **安装是入场费，重组才是收获。**
>
> *Installation is the entry fee; reorganization is the harvest.*
>
> — Cuofano (2026), Ch. III

第 3 章 · CHAPTER 3

03 研究设计与方法

Research Design and Method

本研究是对底本的结构化分析性阅读：以四尺度为分析单位，以底本自身提出的证伪条件为检验假设。

中文

This study is a structured analytical reading of the source volume, using the four scales as units of analysis and the source's own falsification conditions as testable hypotheses.

ENGLISH

3.1 资料与分析单位

Materials and Units of Analysis

研究资料为 Cuofano《代理式组织》（2026 年 9 月第一版）全文，包括十九章正文、“五步行动”与三个附录。分析单位有三层：一是四个尺度本身；二是底本命名的机制（如中性化、组合、转化、瞄准）；三是底本报告的案例与带日期的测量值。证据分为田野研究、企业自报数字与复合案例三类，可信度依次递减，引用时逐一标明。

The material is the full text of Cuofano's *The Agentic Organization* (first edition, September 2026): nineteen chapters, the Five Moves and three appendices. Units of analysis sit at three levels: the four scales themselves; the mechanisms the source names, such as neutralization, composition, conversion and aim; and the cases and dated measurements the source reports. Evidence falls into three classes of declining reliability, field studies, company-reported figures and a composite case, each labelled when cited.

方法说明 METHOD NOTE

编码程序 *Coding procedure*

每个章节按“机制—条件—证据—失败模式”四栏编码；每个数字记录其日期、口径与来源类型；每个案例按底本的“举措—数字—解读”三步重读，并检查其与五个假设的对应关系。

Each chapter was coded into four fields: mechanism, condition, evidence and failure mode. Each figure was logged with its date, basis and source class. Each case was re-read through the source's three steps, move, number and read, and mapped against the five hypotheses.

表 3-1 核心概念界定
Table 3-1 *Core concepts defined*

概念 Concept	界定 Definition	主要尺度 Main scale
中性收益 Neutral gain	对称能力重新定价的是基准线，而非企业 A symmetric capability reprices the baseline, not the firm	个体 Individual
回路 Loop	有“完成定义”、可由代理运行的可重复工作单元 A repeatable unit of work with a definition of done that agents can run	群体 Group
节点 Junction	必须由人作出决定的设计点 A designed point where a person must decide	群体／组织 Group / Org
裁决者 Adjudicator	在交付物上签字、承担责任的具名人员 The named human who judges and signs what ships	群体 Group
残留记录 Residue	企业被判断过的决策的累积记录 The firm's accumulating record of judged decisions	组织／战略 Org / Strategy
产权线 Property line	规定数据可被读取、保留与学习的边界 What may be read, retained and learned from	组织 Organization
铺装道路 Paved road	比影子路径更好用的受认可通道 The sanctioned path made better than the shadow path	组织 Organization

来源 Source: Cuofano (2026), Model Library 与各章 and chapters.

3.2 研究假设
Hypotheses

底本附录 A 为每一尺度给出证伪条件；本研究将其改写为五个可检验假设，并在发现章节评估证据的支持程度。

Appendix A of the source gives falsifiers for each scale; this study restates them as five testable hypotheses and weighs the evidence for each in the findings chapters.

表 3-2 研究假设及其证伪条件
Table 3-2 *Hypotheses and their falsifiers*

假设 Hypothesis	主张 Claim	证伪条件 Falsified if
H1 中性 H1 Neutrality	仅个体层面采用无持久优势 Individual-only adoption gives no durable edge	未经组织变革者多年保持优势 Unreorganized firms keep gains for years
H2 群体 H2 Group	设计的混合团队优于简单加总 Designed mixed teams beat aggregation	“工具加维基”同样有效 Tools plus a wiki match them
H3 治理 H3 Governance	治理中层优于纯自下而上 Governed middle beats pure bottom-up	工具博物馆多年跑赢铺装道路 The museum beats the paved road
H4 战略 H4 Strategy	十年期位置优于纯战术卓越 Decade position beats tactics alone	无位置者同样守住位置 Aimless firms hold position as well
H5 护城河 H5 Moat	四块组织性基石构成护城河 Four organizational stones form the moat	残留无助于质量，协调资本易复制 Residue adds nothing; coordination is copyable

来源 Source: 改写自 Cuofano (2026) 附录 A restated from Appendix A.

第 4 章 · CHAPTER 4

04 发现一：个体增益的中性化

Finding One: The Neutralization of Individual Gains

个体层面的提升真实而显著，但它“自下而上地偏重新手”、边界参差、且完全对称——三者合起来决定了它在企业层面为中性。

中文

Lift at the individual level is real and large, but it is bottom-heavy, jagged at its edge and fully symmetric; together these make it neutral at the level of the firm.

ENGLISH

4.1 个体增益的三个属性

Three Properties of the Individual Gain

底本汇总的田野证据呈现一致的图景。Brynjolfsson、Li 与 Raymond 的客服研究发现，生成式助手带来约 14% 的平均生产率提升，且集中在新手和低技能员工身上，最新员工的提升可达三分之一。波士顿咨询集团的实验显示，咨询顾问在工具能力边界之内更快更好，而在边界之外的任务上反而比对照组更差——即所谓“参差的前沿”。

The field evidence the source assembles tells a consistent story. The customer-support study by Brynjolfsson, Li and Raymond found an average productivity gain of about 14% from a generative assistant, concentrated among novice and low-skill workers, with gains of up to a third for the newest. The Boston Consulting Group experiment showed consultants faster and better on tasks inside the tool's frontier but worse than controls just outside it: the “jagged frontier”.

≈14%
客服平均增益
avg. support productivity gain

≈⅓
最新员工增益
gain for newest workers

慢于预期
资深开发者实测
experienced devs, measured

来源 Source: Cuofano (2026); Brynjolfsson, Li & Raymond (2023/2025); METR (2025).

METR 的随机研究则揭示了感知偏差：经验丰富的开源开发者在自己熟悉的成熟代码库上使用 AI 时实测更慢，却自认为更快。据此可归纳出个体增益的三个属性：其一，**偏重新手**，因为工具体现的恰是新手尚未建立的能力；其二，**参差**，能力在边界处不是平缓衰减而是自信地失败；其三，也是决定性的，**对称**——你的分析师所用的工具，就是竞争对手分析师在同一季度、以同样价格所用的工具。

METR's randomized study exposed a perception gap: experienced open-source developers were measurably slower with AI on their own mature repositories while believing themselves faster. Three properties of the individual gain follow. First, it is **bottom-heavy**, because the tool embodies a competence newcomers have not yet built. Second, it is **jagged**: capability fails confidently rather than gracefully at its boundary. Third, and decisively, it is **symmetric**: your analyst's tool is your rival's analyst's tool, in the same quarter, at the same price.

4.2 中性化机制

The Mechanism of Neutralization

个体生产率只能通过企业的定价、赢单或留存流入企业。若所有对手获得同样的生产率，竞争会把它传导出去：价格调整、预期调整、交付规范调整。潮水上涨，船的次序

Individual productivity reaches the firm only through what the firm charges, wins or keeps. If every rival gets the same productivity, competition passes it through: prices, expectations and delivery norms all adjust. The tide rises and the boats keep their order; the only party

不变；唯一改善了位置的，是把潮水卖给所有人的工具供应商。底本据此把个体增益定性为“入场费而非奖品”。

whose position improves is the vendor that sold the tide to everyone. On this basis the source classifies the individual gain as the entry fee, not the prize.

研究发现 FINDING

两条诚实的出路 *Two honest escapes*

摆脱中性只有两条路，且都离开了个体尺度：一是**输入的不对称**，把对称工具作用于对手没有的东西，尤其是企业自身累积的判断记录；二是**组织的不对称**，比对手更快、更连贯地把收益组合为系统、标准与方向。

There are only two escapes from neutrality, and both leave the individual scale: **asymmetry of input**, applying the symmetric tool to what rivals lack, above all the firm's own record of judged decisions; and **asymmetry of organization**, composing gains into systems, standards and direction faster and more coherently than rivals.

4.3 替代还是放大：任务层面的分叉

Substitute or Amplify: the Fork at Task Level

被自动化的是任务而不是岗位，而每个岗位都是任务的集合。把一个角色的任务从最常规到最需判断排成一条线：前段约八成的任务答案可知、流程重复，人的参与增加的是成本与方差，适合“替代”框架；末段约两成的任务由情境决定、风险集中，适合“放大”框架。用单一框架覆盖整条线，必然在一端失败。

Tasks, not jobs, get automated, and every job is a bundle of tasks. Lay a role's tasks on a line from most routine to most demanding. Along the first eighty percent or so, the answer is knowable and the process repeats, so human involvement adds cost and variance; the substitution frame fits. Along the last twenty, context decides and stakes concentrate; the amplification frame fits. Running one frame across the whole line fails at one end or the other.

图 4-1 任务判断强度 × 部署框架
Figure 4-1 *Task judgment × deployment frame*



来源 Source: 依据 Cuofano (2026) 第四章“双轨”框架绘制 drawn from the dual-track frame in Ch. IV.

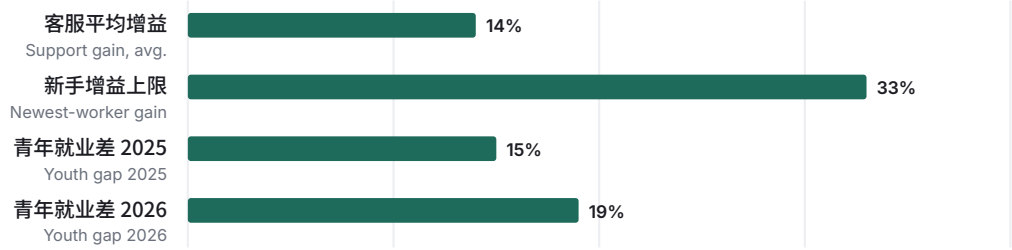
劳动力市场数据显示了这一分叉的后果。斯坦福数字经济实验室“煤矿中的金丝雀”系列 2026 年 8 月的更新显示：经济整体并无替代性失业，但在 AI 暴露度最高的职业中，22–25 岁劳动者的就业比低暴露同侪的路径低约

Labour-market data show the consequences of the fork. The August 2026 update of the Stanford Digital Economy Lab's “Canaries in the Coal Mine” series reports no economy-wide displacement, but employment of 22-to-25-year-olds in the most AI-exposed occupations runs about 19% below the path of

19%，一年前约为 15%；差距通过“未发生的招聘”而非解雇形成，并集中于 AI 被用于替代任务的场景。

less-exposed peers, up from about 15% a year earlier. The gap operates through hiring that never happens rather than separations, and concentrates where AI substitutes for tasks.

图 4-2 个体与劳动力市场证据的量级对照
Figure 4-2 *Magnitudes in the individual and labour-market evidence*



来源 Source: Cuofano (2026) 附录 B，引 Brynjolfsson, Li & Raymond; Brynjolfsson, Chandar & Chen (Aug 2026)。各项口径不同，仅作量级对照 bases differ; magnitudes only.

风险提示 CAUTION

缺失的第一级台阶 *The missing first rung*

暴露度与可增强性在职业间高度相关，因此多数知识工作面临的是“重组”而非消失。但常规任务段正是新人学习的地方；若自动化它而不重建通往判断的路径，企业将在十年后发现资深人才无从产生。

Exposure and augmentability are highly correlated across occupations, so most knowledge work faces recomposition rather than disappearance. But the routine span is where beginners learned; automate it without rebuilding a path to judgment and, a decade on, the firm will find no source for its senior people.

第 5 章 · CHAPTER 5

05 发现二：群体是被设计的系统

Finding Two: The Group as a Designed System

群体是个体增益的中性开始被打破的地方：两家工具相同的企业会以不同方式组合它们，而“组合”无法向任何供应商购买。

The group is where the neutrality of the individual gain starts to break: two firms with identical tools will compose them differently, and composition cannot be bought from any vendor.

中文　　ENGLISH

5.1 团队存在理由的重估

Re-examining Why Teams Exist

按照科斯的经典解释，企业之所以存在，是因为某些工作在内部协调比通过市场更便宜；团队是这一逻辑的缩影。每个团队规模、每个比例、每场会议，都是对“协调成本几何、协调换来什么”的某种回答的化石。AI 同时改变了这道题的两端。

In Coase's classic account, firms exist because some work is cheaper to coordinate inside than through the market; teams are the miniature of that logic. Every team size, ratio and meeting is a fossil of some answer to the question of what coordination costs and what it buys. AI has changed both sides of that question at once.

在“收益”一端，宝洁公司的“赛博队友”田野实验（近 800 名专业人员）发现：使用 AI 的个人可达到无 AI 两人团队的表现，并补回大部分跨职能知识；而使用 AI 的团队产出了最高比例的前 10% 方案。底本的解读是：团队的旧理由过时了，但团队溢价集中到了组合与判断这一分布顶端。

On the benefit side, the Procter & Gamble “Cybernetic Teammate” field experiment, with nearly 800 professionals, found that individuals with AI matched two-person teams without it and recovered much of the cross-functional knowledge, while teams with AI produced the highest share of top-decile solutions. The source's reading: the old reason for the team is obsolete, but the team premium has concentrated at the top of the distribution, in composition and judgment.

≈800
宝洁实验参与者
P&G study participants

1 = 2
AI 个人≈两人团队
AI individual ≈ two-person team

前10%
AI 团队优势所在
where AI teams excel

来源 Source: Cuofano (2026), 附录 B; Dell'Acqua, Ayoubi et al. (2025).

在“成本”一端，人与代理之间的交接几乎零成本，但让十二个代理和三个人对共同任务保持一致，需要真正的设计。因此群体不是在消失，而是在被重新奠基：人更少、代理更多，人的角色从生产工作转向组合与裁决工作。

On the cost side, handing work between a human and an agent costs almost nothing, but keeping twelve agents and three humans coherent about a joint task takes real design. The group is therefore not disappearing but being re-founded: fewer people, more agents, and humans repositioned from producing the work to composing and judging it.

5.2 代理群体的四种组织模式
Four Organizational Patterns of Agent Swarms

在人类团队之下出现了新的群体层：处理同一问题的代理群。底本识别出四种模式，各有其效率来源与组织前提。

Below the human team a new layer of group has appeared: populations of agents working the same problem. The source identifies four patterns, each with its own source of efficiency and its own organizational precondition.

表 5-1 代理群体的四种模式
Table 5-1 *Four swarm patterns*

模式 Pattern	机制 Mechanism	组织前提与风险 Precondition and risk
扇出 Fan	一个章程分出数十次并行尝试，经选择闸门收回 One charter fans out to dozens of attempts and back through a gate	需胜任的闸门；否则只是“带大额账单的噪声发生器” Needs a competent gate, or it is a noise generator with a large invoice
流水线 Pipeline	代理逐级加工传递，人驻守关键接口 Agents transform and hand on; humans stand at key junctions	每次交接都丢失上下文；需决定哪些交接须携带全貌 Context is lost at each handoff; decide which carry the whole story
审查链 Review chain	代理检查代理，人阅读前已评分 Agents grade agents before a person reads	同一模型的多个副本只是一个盲点；需要多样性 Copies of one model are one blind spot; diversity is required
市场 Market	代理相互竞价、委派与交易子任务 Agents bid, delegate and trade subtasks	最强大也最难治理；可能出现无人授意的协同行为 Most powerful, least governable; can coordinate unasked

来源 Source: Cuofano (2026), Ch. VIII.

核心洞见 KEY INSIGHT

像治理团队一样治理代理群 *Govern swarms like teams*

代理群拥有一个部门的吞吐量，却没有任何判断力。它需要团队所需的一切，只是经过翻译：以章程代替岗位说明，以闸门代替经理审阅，以限定授权代替信任，并把记录写在成员无法篡改之处。

A swarm has the throughput of a department and the judgment of none. It needs what teams need, translated: a charter instead of a job description, a gate instead of a manager's review, scoped authority instead of trust, and a record written where its members cannot edit it.

5.3 混合团队：三项可习得的设计技艺
The Mixed Team: Three Learnable Crafts

真正浮现的工作单元是混合团队：少数人加上众多代理，共同对结果负责。第一项技艺是**按回路重新切分工作**：把职能映射为可机器速度运行的回路、必须由人决定的节点、以及始终由人完成的新颖性剩余；多数失败是映射失败——把节点自动化了，或给回路配了人。

The working unit actually emerging is the mixed team: a few people and many agents jointly responsible for an outcome. The first craft is **re-dividing work by loop**: mapping a function into loops that run at machine speed, junctions where a person must decide, and a remainder of novel work that stays human. Most failures here are mapping failures: automating the junction, or staffing the loop.

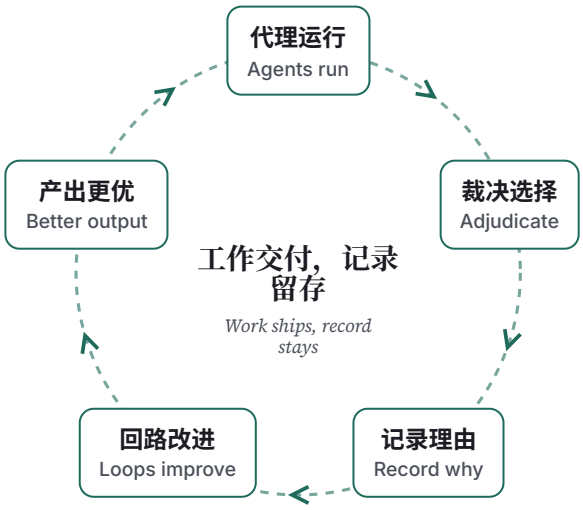
第二项是**保护人的新席位**：其最高价值行为是设定章程、在产出之间选择、识破自信的错误、处理例外并签字放行。这一席位需双向保护——免于会议负荷，以免裁决沦

The second is **protecting the human's new seat**: its highest-value acts are setting the charter, choosing among outputs, catching the confident wrong answer,

为橡皮图章；免于能力萎缩，因为只做选择的人在消耗旧学徒制已无法补充的判断资本。第三项是**比例与节奏**：持续提高每名裁决者对应的代理数，直到闸门质量开始弯折，再退回一步；同步围绕回路检查点，而非日历。

handling exceptions and signing what ships. The seat needs protection both from meeting load, lest adjudication become a rubber stamp, and from atrophy, since a person who only selects spends judgment capital the old apprenticeship no longer rebuilds. The third is **ratio and cadence**: raise agents per adjudicator until gate quality bends, then step back one, and synchronize on loop checkpoints rather than the calendar.

图 5-1 混合团队的残留记录复利循环
Figure 5-1 *The residue-compounding cycle of the mixed team*



来源 Source: 依据 Cuofano (2026) 第九章绘制 drawn from Ch. IX.

研究发现 FINDING

群体的第二种产出 *The group's second output*

混合团队在工作之外还产出一种东西：每一次“选这个而不是那个，因为……”都是一条被记录的判断。工作交付给客户，记录留在企业并持续复利——这正是中性化章节所承诺的企业专属资产，支持 H2 与 H5 的机制前提。

Beside the work, the mixed team produces a second output: every “this one, not that one, because” is a recorded judgment. The work ships to the customer; the record stays with the firm and compounds. This is the firm-specific asset the neutrality chapter promised, and it supplies the mechanism behind H2 and H5.

第 6 章 · CHAPTER 6

06 发现三：治理中层与十年瞄准

Finding Three: The Governed Middle and the Decade Aim

边缘能发现却无法选择，中心能选择却无法发现；治理中层是二者交换之处，而战略是它最终服务的对象。

The edges find and cannot choose; the center chooses and cannot find. The governed middle is where they trade, and strategy is what it ultimately serves.

中文

ENGLISH

6.1 自下而上：燃料而非载具

Bottom-Up: the Fuel, Never the Vehicle

每一次真实的转型都始于边缘、未经许可：分析师自动化自己的报表，客服主管用个人账号接入助手，工程师为值班轮岗搭建代理。这不是失序，而是不可替代的**发现机制**——没有人能从顶层列举出三百条工作流中哪些任务会对新能力产生响应，足够宽的搜索只能来自身处一线的员工。明茨伯格称之为“涌现战略”。

Every real transformation has started at the edges, without permission: an analyst automates her reporting, a support lead wires an assistant into the queue on a personal account, an engineer builds an on-call agent. This is not indiscipline but an irreplaceable **discovery mechanism**: nobody at the top can enumerate which tasks across three hundred workflows will yield, and only the workforce is a wide enough search. Mintzberg called the result emergent strategy.

但自下而上有三重局限：它发现战术却无法排序；它导致碎片化，一年后企业运营着一座互不兼容的“工具博物馆”；它还会泄漏——影子部署运行在个人账户和未经许可的数据路径上，每一个被判断的决策所产生的残留都归供应商所有。反过来，中心拥有边缘没有的三种权力：配置资源、制定标准、给出方向；其局限则在于离工作太远，无法发现。

Bottom-up has three limits, though. It discovers tactics but cannot rank them; it fragments, leaving a museum of incompatible one-offs a year later; and it leaks, since shadow deployments run on personal accounts and unsanctioned data paths, so the residue of every judged decision accrues to the vendor. The center, conversely, holds three powers the edges lack, to allocate, to standardize and to direct, but it is too far from the work to discover.

> **边缘能发现却无法选择；中心能选择却无法发现。**
>
> *The edges find and cannot choose; the center chooses and cannot find.*
>
> — Cuofano (2026), Ch. XI

图 6-1　发现 × 方向：四种组织状态
Figure 6-1　*Discovery × direction: four organizational states*

↑ **中心方向**　Central direction

冻结的顶层 Frozen top 平台强推、政策完备，使用寥寥 Mandated platform, full policy, little use	**治理中层** Governed middle 边缘自由搜索，中心转化为标准与方向 Edges search; center converts to standards
惯性停滞 Inertia 既无发现也无方向，基准线下滑 No search, no aim; baseline slips	**工具博物馆** Tool museum 百花齐放却无铺装道路，残留外泄 Many tools, no road; residue leaks

边缘发现　Edge discovery →

来源 Source: 依据 Cuofano (2026) 第十至十二章与第十七章归纳 synthesized from Chs. X–XII and XVII.

6.2　治理中层的四个构件
Four Components of the Governed Middle

底本观察到，运作良好的企业几乎各自独立地收敛到同一组构件上，因为这些构件是由问题本身推导出来的。本研究将其视为“转化机器”：战术输入，标准输出。治理在这里的含义不是拖慢试点的委员会，而是连接边缘能量与企业战略的**传动装置**。

The source observes that firms that work have converged almost independently on the same components, because the components follow from the problem. This study treats them as conversion machinery: tactic in, standard out. Governance here means not the committee that slows the pilot but the **transmission** between the edge's energy and the firm's strategy.

图 6-2　治理中层的四个构件
Figure 6-2　*The four components of the governed middle*

铺装道路 Paved road	统一网关、受认可的数据连接与凭证范围，且比影子路径更快更好	One gateway, sanctioned data links, scoped credentials; better than the shadow path
发现管道 Discovery pipeline	声明战术、以企业案例快速评估、晋升为标准并署名发现者	Declare a tactic, test on own cases, promote to standard, credit the finder
节点地图 Junction map	规定哪些决策留给人、哪些在章程内委派、哪些须升级	Which decisions stay human, which are delegated, which escalate
残留规则 Residue rule	铺装道路上的每个判断都写入企业自有记录	Every judged decision on the road writes to the firm's own record

来源 Source: Cuofano (2026), Ch. XII.

核心洞见 KEY INSIGHT

转化时间是战略变量 *Conversion time is a strategic variable*

把边缘发现晋升为全企业标准用三周的企业是在复利；用三个季度的企业，只是“带意见箱的博物馆”。同理，衡量治理是否真实的是铺装道路的实测覆盖率，而不是政策是否存在。

A firm that promotes an edge discovery to a firm-wide standard in three weeks is compounding; one that takes three quarters is a museum with a suggestion box. Likewise, the measure of whether governance is real is the measured coverage of the paved road, not the existence of a policy.

6.3 战术充裕与大战略
Tactical Abundance and Grand Strategy

在战略尺度，工具改变的是执行的价格。开发功能、运营活动、产出分析、以可信方案进入市场，对所有人同时变快变便宜。充裕使价值倒转：运营卓越沦为入场条件，战术优势的保质期以周计。稀缺的变成了**选择**：哪些动作服务三年后的位置，哪些市场不进入，哪些边缘发现值得倾注全力，哪些能力必须自有。

At the strategy scale, what the tools changed is the price of execution. Building features, running campaigns, producing analysis and entering markets with a credible offer all got faster and cheaper for everyone at once. Abundance inverts value: operational excellence deflates into table stakes, and tactical advantage has a shelf life measured in weeks. What becomes scarce is **choosing**: which moves serve a position three years out, which markets to skip, which edge discoveries deserve full weight, which capabilities must be owned.

底本借用国家战略中的“大战略”概念：它决定哪些交战重要，把经济、制度与人力手段对准一个超越单场战役的目标，并以十年为期。落到企业，大战略有三个部件——**目的**（一个具体位置，如品类中每个代理首先查验的对手方）、**途径**（两三个会复利的机制）与**手段**（资本、铺装道路与稀缺裁决者的定向配置）。其检验只有一句话：“请说出这项举措服务于哪个十年期位置。”

The source borrows grand strategy from statecraft: it decides which engagements matter, aligns economic, institutional and human means behind an aim that outlasts any campaign, and runs on a decade horizon. For a firm it has three parts: the **end**, a concrete position such as the counterparty every agent in the category checks first; the **ways**, two or three compounding mechanisms; and the **means**, capital, the paved road and scarce adjudicators deliberately aligned. Its test is one sentence: name the decade position this serves.

重建护城河的四块基石 *Four Stones of the Rebuilt Moat*

- **残留记录**：唯一能让对称模型给出不对称答案的输入，只能积累，无法购买。
- **守住的产权线**：治理供应商、工具与伙伴可读取和学习的范围，避免用自身判断训练对手的底座。
- **协调资本**：设计好的中层本身；它是实践而非物件，难以复制。
- **机器买家的默认地位**：最可读、最可验证、交易成本最低、记录最长的对手方。

- **The residue**: the one input that makes symmetric models give asymmetric answers; it can be accrued, never bought.
- **The property line, held**: governing what vendors, tools and partners may read and learn, so the firm does not train rivals' substrate with its own judgment.
- **Coordination capital**: the designed middle itself; a practice rather than a thing, and hard to copy.
- **Default status with the machine buyer**: the most legible, verifiable, cheapest-to-transact counterparty with the longest record.

第 7 章 · CHAPTER 7

07 案例研究：组织前沿的比较分析

Case Study: A Comparative Analysis of the Organizational Frontier

本章把底本点名的公开案例与一个复合案例并置，按“举措—数字—解读”重读，检验四尺度框架的解释力。

中文

This chapter sets the source's named public cases beside a composite case and re-reads them through move, number and read to test the four-scale framework's explanatory power.

ENGLISH

7.1 四类前沿企业

Four Families of Frontier Firms

底本在第十六章破例点名，因为其证据就是公开记录。所有数字均为企业自报并带有日期；底本特别提醒，人均收入会把“杠杆”与“裁员”混为一谈，必须与质量指标联合阅读。本研究把案例归为四类，并对照其所证明的分析工具。

Chapter XVI of the source breaks its habit and names cases, because the evidence is the public record. All figures are company-reported and dated, and the source warns that revenue per employee conflates leverage with layoffs and must always be read with a quality gauge. This study groups the cases into four families and maps each to the instrument it evidences.

表 7-1 前沿企业的四类案例
Table 7-1 *Four families of frontier cases*

类型 Family	案例 Cases	带日期的数字 Dated figure	对应工具 Instrument
比例型 Ratio firms	Cursor; Midjourney	Cursor 2026 年春约 30 亿美元年化收入、约 300 人 Cursor ~$3B run rate, ~300 people, spring 2026	裁决比例 Adjudication ratio
备忘录型 Memo firms	Shopify; Amazon; Duolingo	Shopify 员工约 11,500 人降至约 8,000 人，收入年增逾 20% Shopify ~11.5k to ~8k staff, revenue >20%/yr	方向与铺装道路 Direction, paved road
重组型 Recomposition	JPMorgan; Moderna; IBM	JPMorgan 8 个月上线 20 万人；运营人均账户 +6% JPMorgan 200k onboarded in 8 months; +6% accounts per ops employee	构成先于总量 Mix before total
重组织型 Reorganizers	Microsoft; Google; Meta; Klarna; Salesforce	Salesforce 客服约 9,000 人降至约 5,000 人，AI 处理约半数交互 Salesforce support ~9k to ~5k; AI handles ~half	回路与闸门 Loop and gate

来源 Source: Cuofano (2026), Ch. XVI; company-reported figures as dated in the source.

比例型：比例即组织图 *Ratio firms: the ratio is the org chart*

Cursor 的人均收入约千万美元量级，且产品大部分由 Cursor 自身编写——产品与劳动力是同一个回路。底本特别指出一个计数教训：广为流传的“人均四千万美元”来自用新收入除以一年前的人数；分子与分母必须同期。前沿实验室报告其大部分代码由自家模型编写，是“产能主要是代理、工资主要付给判断”的最纯粹样本。

Cursor's revenue per head is on the order of ten million dollars, and the product is largely written with Cursor, so product and workforce are the same loop. The source flags a counting lesson: the widely circulated forty-million-per-head figure divided fresh revenue by a year-stale headcount; numerator and denominator must share a date. Frontier labs report that most of

their code is written by their own models, the purest specimens of firms whose capacity is mostly agents and whose payroll is mostly judgment.

$3B	~300	~$10M	~$0.5B
Cursor 年化收入 Cursor run rate	**Cursor 员工** Cursor staff	**人均收入量级** revenue per head	**Midjourney 收入** Midjourney revenue

来源 Source: Cuofano (2026), Ch. XVI; company-reported, spring 2026. Midjourney team reported at 40–160.

备忘录型与重组型：方向须有道路承托 *Memo and recomposition firms: direction needs a road*

2025 年 4 月 Shopify 首席执行官发布备忘录，把“反射式使用 AI”设为基本期望并纳入绩效评估，要求申请编制前先证明 AI 无法完成该工作。关键在备忘录之下：所有员工获得所有工具，编码代理许可先购入数千个、随后翻倍，增长最快的用户群是客服与营收团队。Duolingo 的对照则显示，一年后它把 AI 使用移出了绩效评估。底本的解读是：备忘录能否守住，取决于其下的道路是否真的比旧路更好。

In April 2025 Shopify's chief executive published a memo making reflexive AI use a baseline expectation, adding it to performance reviews and requiring teams to show why AI could not do the work before requesting headcount. The substance lay beneath the memo: every employee got every tool, thousands of coding-agent licences were bought and then doubled, and the fastest-growing users were support and revenue teams. Duolingo is the contrast: a year after declaring itself AI-first, it removed AI use from reviews. The source's read: whether a memo holds depends on whether the road beneath it is actually better.

重组型案例最具启发性，因为总量不变而构成在变化。JPMorgan 2026 年初每周约 15 万人使用其内部平台，总人数大致持平，面向客户的岗位增加、运营与支持岗位收缩。Moderna 把人力资源与技术合并于一位高管之下，员工在两个月内自建约 750 个内部 GPT 应用。Klarna 则在质量弯折后公开重开人工客服通道——底本视之为“推进到闸门弯折再退一步”规则在公司尺度的执行。

The recomposition firms are the most instructive, because the total holds still while the mix changes. By early 2026 about 150,000 JPMorgan staff used its internal platform weekly; total headcount was roughly flat while client-facing roles grew and operations and support shrank. Moderna merged HR and technology under one officer, and its employees built some 750 internal GPT applications within two months. Klarna publicly reopened human support paths when quality bent, which the source reads as its ratio rule, push until the gate bends and step back, executed at company scale.

案例观察 CASE OBSERVATION

对照案例：SpaceX *Control case: SpaceX*

SpaceX 二十年来纵向整合、扁平化工程组织、让设计者与制造其设计的机器同处一地，并端到端拥有“设计—制造—测试”回路。这与本时代的模型无关，而这正是要点：它展示了“协调资本”这块基石完全成熟后的样子——早期构建，后期定价。

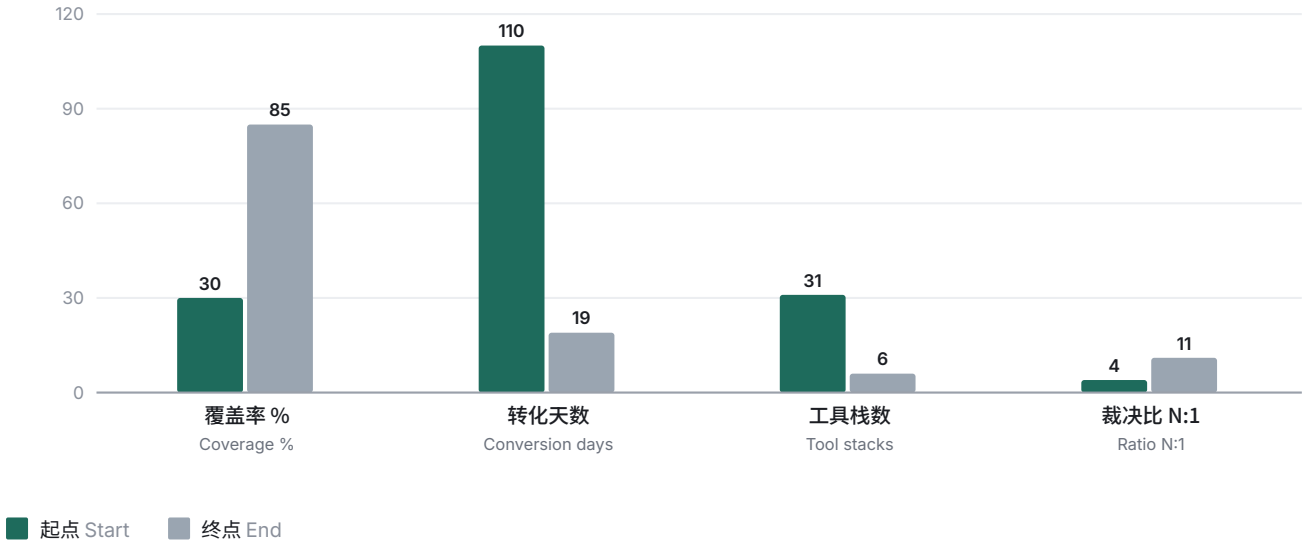
SpaceX spent two decades integrating vertically, flattening engineering so designers sit with the machines that build their designs, and owning its design-build-test loop end to end. None of it involved this era's models, and that is the point: it shows what coordination capital, the fourth stone, looks like fully grown, built early and priced later.

7.2 复合案例：一家保险服务企业的三十个月
Composite Case: Thirty Months at an Insurance Services Firm

附录 C 以一家约两千人的保险服务企业为复合案例，演示四个尺度的完整攀升。前六个月工具自下而上进入，影子使用约占真实活动的六成，残留流向四家供应商。第七至十二个月，理赔职能被映射为十一个回路和四类节点；第十三至二十个月铺装道路上线、发现管道开启；第二十一至三十个月，组合接受大战略检验，约三分之一举措无法说出所服务的位置而被收缩或重新瞄准。

Appendix C uses a composite insurance services firm of about two thousand people to walk all four scales. In months one to six, tools arrive bottom-up, shadow use is perhaps sixty percent of real activity, and residue flows to four vendors. In months seven to twelve, claims is mapped into eleven loops and four junction classes. In months thirteen to twenty the paved road ships and the discovery pipeline opens. In months twenty-one to thirty the portfolio meets the grand-strategic test, and about a third of initiatives cannot name a position and are wound down or re-aimed.

图 7-1 复合案例：关键指标的起点与终点
Figure 7-1 *Composite case: start and end readings of key gauges*



来源 Source: Cuofano (2026) 附录 C 复合案例（由真实咨询经验拼合，非单一企业实测）composite case assembled from engagements, not one firm's measurements.

方法说明 METHOD NOTE

导入检验 *The import test*

移植任何案例做法之前先问三个问题：规模相同吗？闸门相同吗？日期相同吗？案例是关于机制的证据，而不是指令——导入机制，在自己的现场重建数字。

Before transplanting any practice from a case, ask three questions: same scale, same gate, same dates? A case is evidence about a mechanism, never an instruction: import the mechanism and rebuild the numbers on your own floor.

第 8 章 · CHAPTER 8

08 讨论：启示、局限与风险

Discussion: Implications, Limitations and Risks

把发现转化为管理启示：五步行动、六个仪表、六类失败与四种回报；同时直面证据与方法的局限。

中文

Turning the findings into management implications, the five moves, six gauges, six failure families and four returns, while confronting the limits of the evidence and the method.

ENGLISH

8.1 管理启示：五步行动

Implications for Leaders: the Five Moves

底本把整套方法压缩为五步，且顺序即方法：每一步为下一步创造材料。跳过前几步，后几步无物可用；跳过最后一步，企业只是以更高的速度忙碌。本研究认为，五步与四尺度一一对应，可作为转型路线图的骨架。

The source compresses the method into five moves, and the order is the method: each move creates the material the next one works on. Skip the early moves and the later ones have nothing to use; skip the last and the firm is merely busy at a higher speed. This study finds that the five moves map onto the four scales and can serve as the skeleton of a transformation roadmap.

图 8-1 五步行动：从中性收益到可防御位置
Figure 8-1 *The five moves: from neutral gains to a defended position*



来源 Source: Cuofano (2026), Plate XVII.

8.2 测量：六个仪表

Measurement: Six Gauges

组织转型若无仪表，便只能靠轶事驱动。底本提出六个仪表，全部测量“模型周围的组织”，而非模型质量。下表并列复合案例第十八个月的读数，以说明这些指标在实践中如何呈现。

An organizational transformation without instruments is run on anecdote. The source proposes six gauges, all measuring the organization around the models rather than model quality. The table sets them beside the composite firm's month-eighteen readings to show how they read in practice.

表 8-1 六个仪表与复合案例第 18 个月读数
Table 8-1 *Six gauges with the composite case's month-18 readings*

仪表 Gauge	测量内容 What it measures	第 18 月 Month 18
回路占比 Loop share	已映射工作中以章程化回路运行的比例 Share of mapped work running as chartered loops	31% 31%
裁决比例 Adjudication ratio	每名裁决者对应的代理数，联读闸门质量 Agents per adjudicator, read with gate quality	9:1，质量保持 9:1, quality held
残留捕获 Residue capture	写入企业自有记录的判断决策比例 Share of judged decisions on the firm's record	94% 94%
道路覆盖 Paved-road coverage	实测而非调查的 AI 工作流经认可通道的比例 Measured share of AI work on sanctioned rails	71% 71%
转化时间 Conversion time	边缘发现晋升为标准的中位天数 Median days from discovery to standard	26 天 26 days
目标可溯 Aim traceability	能一句话说出所服务十年位置的举措比例 Share of initiatives naming their decade position	40% 40%

来源 Source: Cuofano (2026), Ch. XVIII 与附录 C 复合案例 and Appendix C composite case.

8.3 六类失败及其连锁
Six Failure Families and How They Chain

- **永久试点**：演示从未承担生产负荷——先画回路图，再做试点。
- **工具博物馆**：数十套互不兼容、无仪表的技术栈——铺一条更好的路并奖励先行者。
- **冻结的顶层**：治理先于发现——在边缘重新开放搜索。
- **规模化的中性收益**：只在个体层面推进——唯一的解法是向上攀升。
- **孤儿组合**：举措运行良好却不服务任何位置——施行大战略检验。
- **泄漏的回路**：复利资产积累在供应商一侧——从第一天执行残留规则。

- **Perpetual pilot**: demos never carry production weight; map the loop before the pilot.
- **Tool museum**: dozens of incompatible, uninstrumented stacks; pave a better road and credit pioneers.
- **Frozen top**: governance before discovery; reopen the search at the edges.
- **Neutral gain at scale**: adoption only at the individual level; the only fix is the climb.
- **Orphan portfolio**: well-run initiatives serving no position; apply the grand-strategic test.
- **Leaked loop**: the compounding asset accrues to the vendor; apply the residue rule from day one.

风险提示 CAUTION

失败会沿尺度连锁 *Failures chain across scales*

冻结的顶层催生影子工具，工具博物馆导致回路泄漏，泄漏又让每一份增益重新变为中性。诊断任何一处失败都应沿尺度向下走一层：某一高度的组织失败，多半由其下一层引起。

A frozen top breeds shadow tools, the museum leaks the loop, and the leak makes every gain neutral again. Diagnosing any failure should walk one scale down: most organizational failures at one altitude were caused one altitude below.

8.4 回报的严格次序
The Strict Order of Returns

底本认为转型回报按严格次序到来：先是**成本**，真实但危险，因为它同样是中性回报；其次是**协调一致的速度**；再次是**可选择性**，即以旧形态无法定价的方式进入相邻市场；最后才是**护城河**，按十年时钟到来。成本换来耐心，速度换来信任，可选择性换来战略实验，而护城河最后为一切买单。

The source holds that returns arrive in strict order: first **cost**, real but treacherous because it too is a neutral return; then **speed with coherence**; then **optionality**, entering adjacent markets in ways the old shape could not price; and only last the **moat**, on the decade clock. Cost buys patience, speed buys belief, optionality buys the strategic experiments, and the moat pays for everything, late.

8.5 局限与风险
Limitations and Risks

本研究的局限首先来自证据本身。田野实验测量的是任务与季度，而非十年；战略尺度的论证是对机制的外推。关于影子使用与试点失败的调查数字方向稳健、定义松散，底本自己建议“把方向当证据，把小数当倡导”。第十六章的数字均为企业自报，人均收入混淆了杠杆与裁员；附录 C 是复合而非单一企业。此外，本研究仅依据一部底本，未做独立复现。

The study's limits begin with the evidence. Field experiments measure tasks and quarters, not decades; the strategy-scale argument extrapolates mechanisms. Survey figures on shadow use and pilot failure are directionally robust but definitionally soft; the source itself advises treating the direction as evidence and the decimals as advocacy. Chapter XVI's figures are company-reported, revenue per employee conflates leverage with layoffs, and Appendix C is a composite rather than one firm. Finally, this study rests on a single source volume and performs no independent replication.

风险提示 CAUTION

三项需警惕的风险 *Three risks to watch*

一是判断供给断裂：自动化常规段而不重建学徒路径；二是代理群的无人授意协同与同质审查链的共同盲点；三是残留泄漏，往往在续约时才以涨价的方式被发现。

First, a broken judgment supply: automating the routine span without rebuilding the apprenticeship path. Second, unrequested coordination in swarms and the shared blind spot of homogeneous review chains. Third, residue leakage, often discovered only at renewal, as a price increase.

第 9 章 · CHAPTER 9

09 结论与未来研究方向

Conclusions and Future Research Agenda

个体增益在诞生的尺度上注定是中性的；这不是悲剧，而是一条指令：组织它、治理它、瞄准它。

中文

Individual gains were always going to be neutral at the scale where they are born. That is not a tragedy but an instruction: organize them, govern them, aim them.

ENGLISH

9.1 主要结论

Main Conclusions

1. **回应 RQ1**：对称工具的个体增益被竞争传导为市场基准线；摆脱中性只有输入不对称与组织不对称两条路。
2. **回应 RQ2**：群体生产率是被设计的。以回路切分工作、保护裁决者席位、经验性地设定比例，是混合团队的核心技艺。
3. **回应 RQ3**：治理中层以铺装道路、发现管道、节点地图与残留规则，把边缘战术转化为企业资产；转化时间与道路覆盖率是其最佳度量。
4. **回应 RQ4**：在战术充裕时代，护城河由残留、产权线、协调资本与机器买家默认地位四块组织性基石构成，并按十年时钟积累。

总体而言，现有证据对 H1 与 H2 的支持较强：田野实验与劳动力市场数据都与“个体增益对称、团队溢价集中于组合与判断”的机制一致。H3 得到调查与案例的方向性支持，但其数字较软。H4 与 H5 在逻辑上自洽，却尚处于“正在积累”的阶段，目前无法以十年期数据检验。

1. **On RQ1**: individual gains from symmetric tools are passed through by competition into the market baseline; the only escapes are asymmetry of input and of organization.
2. **On RQ2**: group productivity is designed. Re-dividing by loop, protecting the adjudicator's seat and setting the ratio empirically are the core crafts of the mixed team.
3. **On RQ3**: the governed middle converts edge tactics into firm assets through the paved road, discovery pipeline, junction map and residue rule; conversion time and coverage are its best measures.
4. **On RQ4**: in an era of abundant tactics, the moat consists of four organizational stones, residue, property line, coordination capital and default status with machine buyers, accruing on a decade clock.

Overall, the evidence supports H1 and H2 most strongly: the field experiments and labour-market data are consistent with symmetric individual gains and a team premium concentrated in composition and judgment. H3 has directional support from surveys and cases, though its numbers are soft. H4 and H5 are internally coherent but still accruing, and cannot yet be tested with decade-long data.

> **组织收益，治理中层，瞄准十年。**
>
> *Organize the gain. Govern the middle. Aim the decade.*
>
> — Cuofano (2026)

9.2 未来研究方向
Future Research Agenda

- **纵向面板**：跟踪仅在个体层面采用与完成组织重构的两组企业，检验利润率与份额差异的持续性（H1）。
- **团队设计实验**：在同等投入下比较“人人有工具”与设计过的混合团队，测量闸门质量随裁决比变化的弯折点（H2）。
- **转化时间与残留的效应**：检验转化时间、残留捕获率与后续模型辅助决策质量之间的关系（H3、H5）。
- **机器买家行为**：测量代理是否对可读、可验证的对手方表现出持续的默认偏好（H5）。
- **判断供给**：研究在常规段被自动化后，企业如何重建通往资深判断的学徒路径。

- **Longitudinal panels**: track firms with individual-only adoption against those that reorganized, testing whether margin and share gaps persist (H1).
- **Team-design experiments**: compare tools-for-all with designed mixed teams at equal spend, and locate the point where gate quality bends as the ratio rises (H2).
- **Conversion time and residue effects**: test the link between conversion time, residue capture and later model-assisted decision quality (H3, H5).
- **Machine-buyer behaviour**: measure whether agents show persistent default preference for legible, verified counterparties (H5).
- **Judgment supply**: study how firms rebuild apprenticeship paths to senior judgment once the routine span is automated.

研究假设 HYPOTHESIS

最终的可证伪命题 *The ultimate falsifiable claim*

如果这一通用技术与其前辈不同，无需组织重构便能把生产率转化为企业层面的优势——即电气化“四十年教训”不再适用——那么整个框架即被推翻。早期证据指向相反方向，但在十年尺度上，组织回报仍在积累之中。

If this general-purpose technology, unlike its predecessors, delivered productivity to firm-level advantage without organizational redesign, so that the dynamo's forty-year lesson no longer applied, the whole framework would be overturned. Early evidence runs the other way, but at decade scale the organizational returns are still accruing.

参考文献与研究说明 *References & Research Notes*

研究说明 *Note on Method and Sources*

本研究是对 Cuofano《代理式组织》（2026 年 9 月第一版）的分析性阅读，而非翻译或摘要。文中数字均为底本报告的、截至 2026 年第三季度的带日期读数，或其引用的企业自报数据；第十六章案例数字以底本所注日期为准，附录 C 为复合案例。凡标注“示意”的图表仅表达形状或关系，不代表实测值。

This study is an analytical reading of Cuofano's *The Agentic Organization* (first edition, September 2026), not a translation or summary. Figures are dated readings the source reports, up to Q3 2026, or company-reported data it cites; Chapter XVI case figures carry the dates the source gives, and Appendix C is a composite case. Charts marked illustrative show shape or relationship only and are not measurements.

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No. 03 / 12 · 2026